



Test Report - TIA-Demo Prototype-001

Test Information

Project: TIA-Demo DIP Configuration
Operator: Tobias-NUC
Date: 09.08.2026
Time: 16:03:28
Duration: 52,04s

Device Under Test

Serial Number: Prototype-001
Device Name: TIA-Demo
Revision: B
Variant: I2C DAC + MCU
Part Number: -

Additional Notes:

DIP switches determine which tests are executed

PASS

Step	Name	Lower Limit	Measured	Upper Limit	Result
2	VIN 12V?	10,8V	12,013V	13,2V	✓
3	U1 VOUT 3V3?	2,97V	3,295V	3,63V	✓
4	D2 Forward Voltage	1,71V	2V	2,09V	✓
5	R1 Voltage Drop	1,17V	1,301V	1,43V	✓
6	D2 Forward Current	1,17mA	1,301mA	1,43mA	✓
10	Segment A ON?	2,97V	3,271V	3,63V	✓
12	Segment B ON?	2,97V	3,27V	3,63V	✓
14	Segment C ON?	2,97V	3,27V	3,63V	✓
16	Segment D ON?	2,97V	3,271V	3,63V	✓
18	Segment E ON?	2,97V	3,27V	3,63V	✓
20	Segment F ON?	2,97V	3,271V	3,63V	✓
22	Segment G ON?	2,97V	3,271V	3,63V	✓
23	### Read DIP Switches ###	0	0	255	✓
24	### DIP Switch 1 ON? ###	0	0	0	✓
27	D3 Red Forward Voltage	1,53V	1,778V	1,87V	✓
28	R22 Voltage Drop	1,35V	1,518V	1,65V	✓
29	D3 Red Forward Current	1,143mA	1,265mA	1,397mA	✓
31	D3 Green Forward Voltage	2,07V	2,309V	2,53V	✓
32	R23 Voltage Drop	900mV	986,438mV	1,1V	✓
33	D3 Green Forward Current	747uA	822,032uA	913uA	✓



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Step	Name	Lower Limit	Measured	Upper Limit	Result
35	D3 Blue Forward Voltage	2,25V	2,534V	2,75V	✓
36	R24 Voltage Drop	675mV	760,199mV	825mV	✓
37	D3 Green Forward Current	576uA	633,499uA	704uA	✓
38	### DIP Switch 2 ON? ###	0	0	0	✓
41	Button 1 High?	2,97V	3,288V	3,63V	✓
42	Button 1 Low?	-200mV	2,426mV	200mV	✓
43	Button 2 High?	2,97V	3,288V	3,63V	✓
44	Button 1 Low?	-200mV	1,918mV	200mV	✓
45	### DIP Switch 3 ON? ###	0	0	0	✓
48	Divider with 680R	189mV	210,412mV	231mV	✓
49	Divider with 1k2	315mV	353,53mV	385mV	✓
50	Divider with 2k2	531mV	593,294mV	649mV	✓
51	Divider with 3k9	828mV	924,196mV	1,012V	✓
52	Divider with 6k8	1,17V	1,333V	1,43V	✓
53	### DIP Switch 4 ON? ###	0	0	0	✓
56	POT1 Adjusted to 1,5V	1,45V	1,5V	1,55V	✓
57	### DIP Switch 5 ON? ###	0	0	0	✓
60	Temperature Sensor VOUT	693mV	730,37mV	847mV	✓
61	Temperature	20,7°C	23,037°C	25,3°C	✓
62	### DIP Switch 6 ON? ###	0	0	0	✓
64	Set I2C DAC to 0.5V	1	1	1	✓
65	Measure DAC Output 0.5V?	450mV	500,009mV	550mV	✓
66	Set I2C DAC to 1.0V	1	1	1	✓
67	Measure DAC Output 1.0V?	900mV	997,661mV	1,1V	✓
68	Set I2C DAC to 1.5V	1	1	1	✓
69	Measure DAC Output 1.5V?	1,35V	1,496V	1,65V	✓
70	Set I2C DAC to 2.0V	1	1	1	✓
71	Measure DAC Output 2.0V?	1,8V	1,993V	2,2V	✓
72	Set I2C DAC to 2.5V	1	1	1	✓
73	Measure DAC Output 2.5V?	2,25V	2,491V	2,75V	✓
74	Set I2C DAC to 3.0V	1	1	1	✓
75	Measure DAC Output 3.0V?	2,7V	2,99V	3,3V	✓
76	### DIP Switch 7 ON? ###	0	0	0	✓



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Step	Name	Lower Limit	Measured	Upper Limit	Result
78	Program MCU	0	0	0	✓
79	Get FW-Version	1,2	1,2	1,2	✓
80	### DIP Switch 8 ON? ###	0	0	0	✓
83	Read POT2 Value from MCU	1450mV	1491mV	1550mV	✓
84	Configure MCU DAC to 0.5V	1	1	1	✓
85	Read DAC Output from MCU ADC	450mV	500mV	550mV	✓
86	Measure DAC Output Voltage	450mV	501,18mV	550mV	✓
87	Configure MCU DAC to 1.0V	1	1	1	✓
88	Read DAC Output from MCU ADC	950mV	995mV	1050mV	✓
89	Measure DAC Output Voltage	950mV	998,212mV	1,05V	✓
90	Configure MCU DAC to 1.5V	1	1	1	✓
91	Read DAC Output from MCU ADC	1450mV	1489mV	1550mV	✓
92	Measure DAC Output Voltage	1,45V	1,497V	1,55V	✓
93	Configure MCU DAC to 2.0V	1	1	1	✓
94	Read DAC Output from MCU ADC	1950mV	1987mV	2050mV	✓
95	Measure DAC Output Voltage	1,95V	1,993V	2,05V	✓
96	Configure MCU DAC to 2.5V	1	1	1	✓
97	Read DAC Output from MCU ADC	2450mV	2485mV	2550mV	✓
98	Measure DAC Output Voltage	2,45V	2,489V	2,55V	✓
99	Configure MCU DAC to 3.0V	1	1	1	✓
100	Read DAC Output from MCU ADC	2950mV	2979mV	3050mV	✓
101	Measure DAC Output Voltage	2,95V	2,985V	3,05V	✓
102	Measure Timer Frequency	4,5Hz	4,924Hz	5,5Hz	✓

Used Equipment

Test Hardware:

Device Name: compact Test Interface Adapter
Serial Number: 2026001
Firmware Version: 0.5.2

Instruments:

Rigol DMM: Rigol Technologies,DM858,DM8A274201911,00.01.00.00.22